



Potential Energy and Conservation of Energy

PE
Work
Types of Forces
Conservation of energy

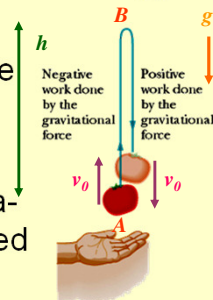
PE and conservation of Energy

- Relationship between Work and change in KE

$$\Delta K = W$$

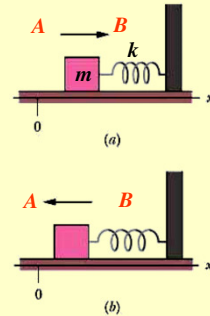
- Throw object up – work done by gravitational force is –ve – energy is transferred from KE of object.
- What happens to this energy?
- Transferred to Gravitational potential energy of the object by the gravitational force.
- Defined **gravitational potential energy**

$$\Delta U = -W$$



PE and conservation of Energy

- For a block-spring system –
 - push the block suddenly toward the right
 - Spring force is towards the left – does negative work on the block
 - Energy transferred from KE of block to **elastic potential energy** of the spring-block system
 - Block slows and eventually stops, and starts to move leftward due to the spring force.
 - Energy transfer is then reversed – from PE of block spring to KE of block.



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Conservative and Non-conservative forces

- Consider a system – 2 or more objects
 - Force acts between an object (tomato or block) and the rest of the system.
 - System configuration changes and force does work W_1 on the object
 - Transfer of energy between KE of object and some other type of energy of the system.
 - When configuration change is reversed, a force reverses the energy transfer – work done is W_2 .

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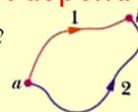
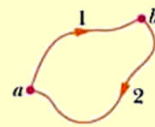
Conservative and Non-conservative forces

- When $W_1 = -W_2$ the other energy is PE and force is a **conservative force** (gravitational force, spring force)
- When a force is a **non-conservative force** KE is transferred to thermal energy (during friction, drag)
- Thermal energy transfer cannot be reversed back to KE

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Path Independence

- A test that will help decide whether a force is conservative or non-conservative.
 - Let the force act on a particle that moves along a closed path $a - b - a$
 - Beginning and ending at same point
 - Force is conservative ONLY IF total energy transfer is **ZERO**
 - $W_{\text{net}} = W_{ab,1} + W_{ba,2} = 0$.
- The net work done by a conservative force on a particle moving around any closed path is zero.
- Gravitational force – conservative
- Test result: The work done by a conservative force on a particle moving between two points does not depend on the path taken by the particle. $W_{ab,1} = -W_{ab,2}$



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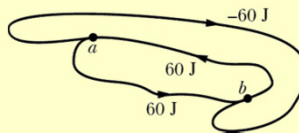
Path independence

- Compare with Ch. 18 in thermodynamics
- There the work done was path dependent – there was a temperature change involved.
- The forces involved were non-conservative

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Checkpoint 1

- The figure shows three paths connecting points a and b . A single force, $\vec{F}$, does the indicated work on a particle moving along each path in the indicated direction. On the basis of this information, is force $\vec{F}$ conservative?



Go through sample problem 8-1

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Question

- Complete the following statement: In situations involving non-conservative external forces, the work done by these forces
 - a) is always negative.
 - b) is always equal to zero.
 - c) is always positive.
 - d) can be either positive or negative.
 - e) usually cannot be determined.

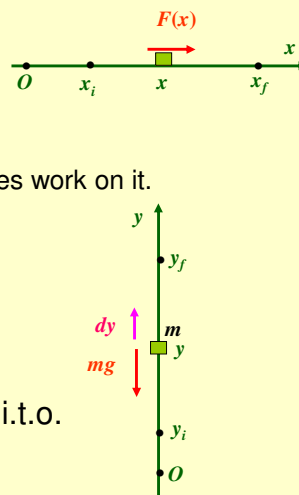
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Determining Potential Energy

- Work done by a force $F(x)$ is $W = \int_{x_i}^{x_f} F(x) dx$
- But $\Delta U = -W$
- Therefore $\Delta U = - \int_{x_i}^{x_f} F(x) dx$
- For gravitational PE:
 - particle moves from y_i to y_f , a force $\vec{F}$ does work on it.
 - Change in gravitational PE:

$$\Delta U = - \int_{y_i}^{y_f} (-mg) dy = mg [y]_{y_i}^{y_f}$$

$$\Delta U = mg(y_f - y_i) = mg\Delta y$$
- Only ΔU has any physical meaning
- But we sometimes want to express U i.t.o. particle height. $\Delta U = U_f - U_i = mg(y - y_i)$
 $U - 0 = U = mg(y - 0) = mgy$



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Determining Potential Energy

- Elastic Potential Energy

- Block-spring system

- Spring constant k , block moves from x_i to x_f

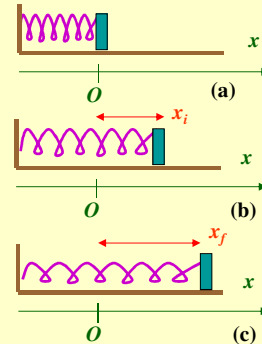
- $F_x = -kx$

- $$\Delta U = - \int_{x_i}^{x_f} (-kx) dx = k \left[\frac{1}{2} x^2 \right]_{x_i}^{x_f}$$

$$\Delta U = \frac{1}{2} k x_f^2 - \frac{1}{2} k x_i^2$$

- With reference at $x = 0$

- $U - 0 = \frac{1}{2} k x^2 - 0 = \frac{1}{2} k x^2$



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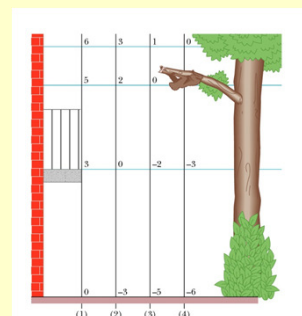
Sample Problem p172

- A 2.0 kg sloth hangs 5.0 m above the ground.

(a) What is the gravitational potential energy U of the sloth-Earth system if we take the reference point $y = 0$ to be

- At the ground,
- At the balcony floor (3 m above the ground)
- At the limb
- 1.0 m above the limb?

(b) The sloth drops to the ground. For each of the reference points, what is the ΔU in the potential energy of the Sloth-Earth system.



(a) Determine U with respect to the reference point:

- $U = mgy = 2.0 \text{ kg}(9.8 \text{ m/s}^2)(5\text{m}) = 98 \text{ J}$

- $U = mgy = 2.0 \text{ kg}(9.8 \text{ m/s}^2)(2\text{m}) = 39 \text{ J}$

- $U = mgy = 2.0 \text{ kg}(9.8 \text{ m/s}^2)(0\text{m}) = 0 \text{ J}$

- $U = mgy = 2.0 \text{ kg}(9.8 \text{ m/s}^2)(-1\text{m}) = -19.6 \text{ J}$

(b) Determine U with respect to the reference point:

- $\Delta U = mg\Delta y = 2.0 \text{ kg}(9.8 \text{ m/s}^2)(-5\text{m}) = -98 \text{ J}$

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Conservation of Mechanical Energy

- Mechanical Energy of an **isolated system** (no external forces) where **only conservative forces** cause energy changes:
- Mechanical Energy: $E_{mech} = K + U$
- But $\Delta K = W$
- And $\Delta U = -W$
- Therefore $\Delta K = -\Delta U$
Decrease in PE = Increase in KE
- When a conservative force is acting on an object

$$\Delta K = -\Delta U$$

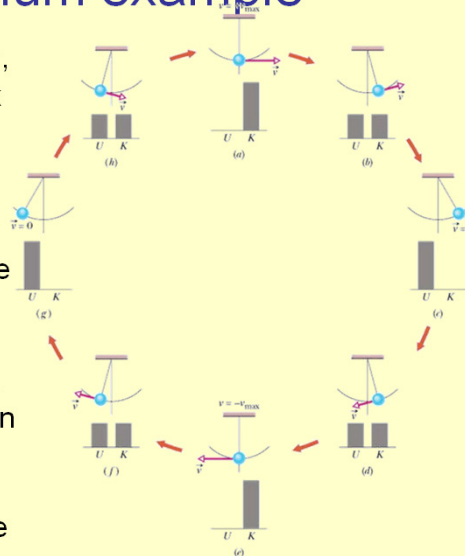
$$K_2 - K_1 = -(U_2 - U_1) = U_1 - U_2$$

$$(K_2 + U_2) = (K_1 + U_1)$$
- **Principle of conservation of Mechanical Energy**

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Pendulum example

- In pendulum-Earth system,
- Energy is transferred back and forth between KE and PE
- **$K + U = \text{constant}$**
- If we know U at highest point, we will know K at the lowest point.
- Reference points
 - lowest point: $U_{min} = 0$ J.
 - Highest point: $v = 0$ then $K_{min} = 0$ J
- $(K_2 + U_2) = (K_1 + U_1)$
- No need to know any force involved.



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Potential Energy Curves

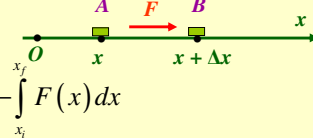
- Particle has a conservative force acting on it
- Particle moves only in x direction
- Force known, determine PE:

$$\Delta U = -W = -\int_{x_i}^{x_f} F(x) dx$$

- PE Known, determine force:

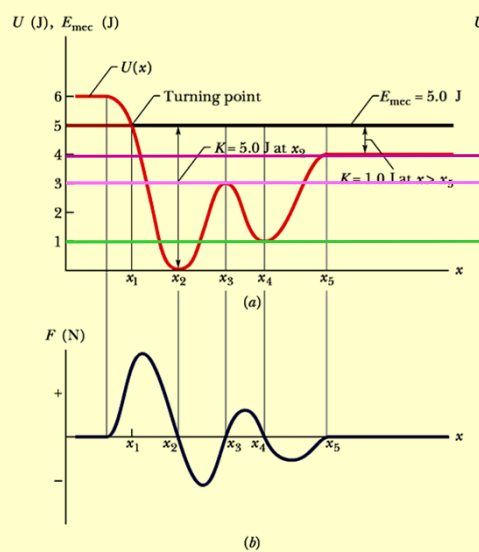
$$F(x) = -\frac{dU}{dx}$$

- Check using $U = \frac{1}{2}kx^2$, or $U = mgx$
- Mechanical Energy: $E = K(x) + U(x)$



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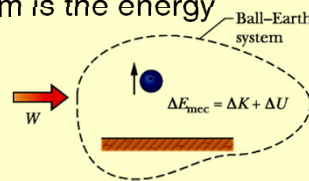
Potential Energy Curves



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Work done by External Force

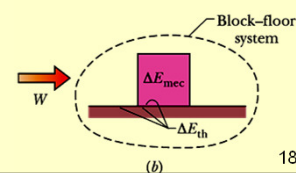
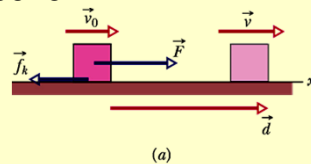
- Def:
 - Work is energy transferred to or from a **system** by means of an external force acting on that system.
- For a single particle – work done by force can only change KE: $\Delta K = W$
- When we push a ball up in the air – external force transfers energy – to what system is the energy transferred?
 - Both ΔK and ΔU are involved
 - $W = \Delta K + \Delta U = \Delta E_{mec}$
 - No friction involved



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Work done by External Force

- Constant force pulls block along x axis through displacement d – velocity increases.
- Kinetic frictional force from the floor on the block.
- System = block (Fig (a))
 - $\sum F_x = F - f_k = ma$
 - $v_x^2 = v_{x0}^2 + 2ad$
 - $Fd = \frac{1}{2}mv^2 - \frac{1}{2}mv_0^2 + f_k d$
 $= \Delta K + f_k d$
 - In general, (when change in PE – e.g. moving up a slope)
 $Fd = \Delta E_{mec} + f_k d = \Delta K + \Delta U + f_k d$
 - $\Delta E_{th} = f_k d$ increase in thermal energy
- System = Block-floor (Fig (b))
 - Work done by external force $W = \Delta E_{mec} + \Delta E_{th}$



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Sample Problem p182

- A food shipper pushes a wood crate of cabbage heads (total mass $m = 14 \text{ kg}$) across a concrete floor with a constant horizontal force of 40 N . In a straight-line displacement of magnitude $d = 0.50 \text{ m}$, the speed of the crate decreases from $v_0 = 0.60 \text{ m/s}$ to $v = 0.20 \text{ m/s}$.
(a) How much work is done by the force and on what system does it do the work?
(b) What is the increase in ΔE_{th} in the thermal energy of the crate and floor?

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Conservation of Energy

Read the section yourself

- In an isolated system, the energy can be transferred from one form to another form, but the **TOTAL ENERGY remains constant**.

$$W = \Delta K + \Delta U + \Delta E_{th} + \Delta E_{int} = 0$$

- If total energy changes (not isolated system)– energy is transferred to or from the system

- $W = \Delta E$

$$= \Delta E_{mech} + \Delta E_{th} + \Delta E_{int}$$

- Power (rate of work done)

$$P = \frac{dE}{dt}$$

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Problem 42

- A worker pushed a 27 kg block 9.2 m along a level floor at constant speed with a force directed 32° below the horizontal. If the coefficient of kinetic friction between block and floor was 0.20, what were (a) the work done by the worker's force and (b) the increase in thermal energy of the block– floor system?

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Example

- A bungee jumper of mass 61 kg jumps of the edge of the bridge. He is attached to an elastic cord which is originally 25 m long. The cord stretches a distance d when the person is hanging below. The spring constant of the cord is 160 N/m. (a) Determine the distance which the cord stretches. (b) Determine the net force on the jumper at the lowest point.

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